25900802 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPERANCE

EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCECT101 - COMPUTER FUNDAMENTALS AND BASICS OF PC HARDWARE

(2024 ADMISION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. Identify the first electronic general-purpose computer. [U] / [CO1]

   a). PDP-1 b). IBM 701

   c). ENIAC d). UNIVAC

2. Identify the largest unit among the following memory capacity units [U] / [CO2]

   a). Megabyte b). Terabyte

   c). Gigabyte d). Kilobyte

3. Identify the maximum data transfer rate for DDR5 memory? [U] / [CO3]

   a). 3,200 MT/s b). 4,800 MT/s

   c). 6,400 MT/s d). 8,400 MT/s

4. Identify the scenario in which we would use hexadecimal representation over binary. [U] / [CO1]

   a). When performing basic arithmetic operations b). For representing characters in a text file

   c). For storage on a hard drive d). For more compact representation of large binary numbers

5. Recognize the type of memory directly accessible by the CPU for quick data retrieval [K] / [CO2]

a). Hard Disk
b). Flash Memory
c). RAM
d). ROM

6. Examine the impact of power consumption when comparing SSDs and HDDs in portable devices. [An] / [CO3]

a). HDDs consume less power, making them ideal for laptops.
b). SSDs consume more power due to their speed.
c). Both consume the same amount of power in portable devices.
d). SSDs use less power, improving battery life in portable devices.

7. Identify the primary characteristic of a LAN. [U] / [CO1]

a). Covers a large geographical area
b). Limited to a small geographic area, typically within a building
c). Uses satellite connections
d). Connects multiple cities

8. Relate why SSD is often preferred for laptops and mobile devices. [U] / [CO2]

a). SSDs are cheaper than HDDs.
b). SSDs have no moving parts, making them more durable and energy efficient.
c). SSDs are slower but more stable than HDDs.
d). SSDs store data magnetically, making them ideal for portability.

9. Identify the primary function of the Central Processing Unit (CPU) from the given list. [U] / [CO1]

a). Data Storage
b). input/ output operations
c). Executing instructions
d). Networking

10. Identify the technology commonly used by modern scanners to scan images. [U] / [CO2]

a). Laser sensor
b). Image sensor (CCD or CIS)
c). Printer head
d). Touchpad

[10x1 = 10]

Part B

Short answer Type Questions

Answer any **4** Questions

Each Question carries **5** marks

11. Describe the components and functioning of a DC regulated power supply with a block diagram. [U] / [CO2]

12. Compare the main differences between inkjet and laser printers. [U] / [CO2]

13. Examine the memory technologies used in GPUs and analyze the different types available. [An] / [CO3]

14. Examine two ways in which Green Computing can help reduce environmental impact. [An] / [CO3]

15. Explain the types and importance of cooling systems in computers. [U] / [CO2]

16. Describe how cache memory improves performance in multi-core processors. [U] / [CO3]

[4x5=20]

Part C

Essay Type Questions

Answer any **2** Questions

Each Question carries **10** marks

17. Discuss the characteristics, advantages, and disadvantages of various types of operating systems [U] / [CO1]

18. Explain the differences between various types of USB ports & CPU Sockets. [U] / [CO2]

19. Examine the concept of neuromorphic computing. [An] / [CO3]

20. Describe the components of a computer system with a simple diagram to support your explanation. [QNSMARTI:T:B20] / [CO1]

[2x10 =20]